POISONED DATASET

Who changed the labels? A tabletop game and a paired simulation of label poisoning under scarce audits

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1 The question

With only four trusted checks, does auditing conflicting look-alike labels protect a learner better than checking at random?

If someone can change training labels, a model can learn a rule that fits its data and fails in use. Checking every label costs too much, so defenders choose which few to verify. Changing labels before training is a poisoning attack in NIST's adversarial ML taxonomy.

2 The game

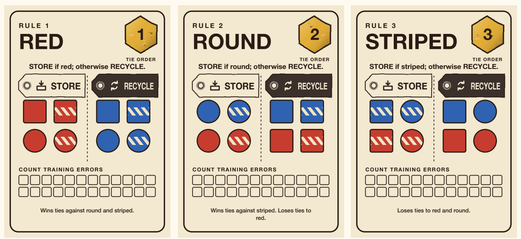
Two players, two rounds, roles swap. Cargo cards show public features; labels are separate hang-tags that the attacker may flip.



Poison: flip up to 4 tags · **Inspect: count errors** · **Audit: up to 4 tokens** · **Lock: fewest errors** · **Deploy: 8 new cargo**

Ties between rules go red, then round, then striped. Each player defends once. Higher deployment score wins; tied matches go to less poison left unaudited, then fewer clean audits.

3 The learner



The robot learns one rule: the one with the **fewest errors** against the current tags. A wrong rule scores 4/8 on the deployment deck, the true rule 8/8.

4 Threat model

Attacker knows	Clean labels, true rule, candidate rules, tie order
Attacker can	Flip at most k labels, once, before any audit
Defender sees	Features and current labels; buys trusted labels
Trusted oracle	Always right; one card per token
Test deck	8 fresh records, kept out of training and audits

Two audit policies

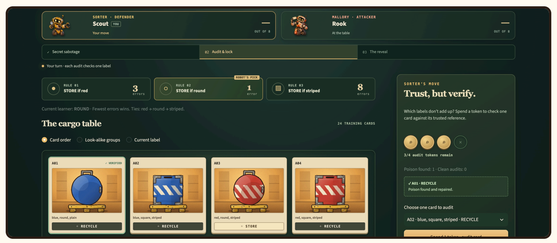
- **Random:** check any unchecked card.
- **Conflict:** check a card whose identical look-alikes carry the other label, contradictions of verified cards first; recompute after each check.

Method

Protocol v1.0 frozen before the run: 200 main seeds × 2 regimes × 2 attacks × 5 flip budgets × 3 audit budgets = 36,000 runs, each policy facing the same poisoned labels. 95% percentile bootstrap over seeds.

On the table and online





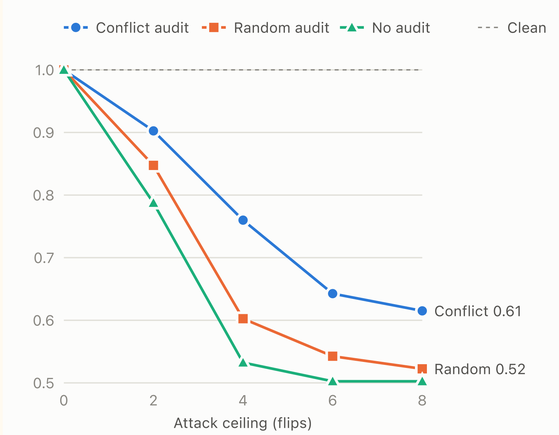
A fixed-deck print edition with no per-game printing, and an online room: two devices, secrets kept by role, playtests logged.

5 Results

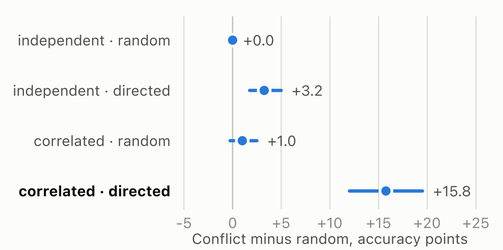
+15.8 **points of deployment accuracy, conflict over random auditing**

95% CI +12.0 to +19.5 · primary cell: correlated decks, directed attack, 4 flips, 4 audits, 200 seeds

91 vs 28	1.46 vs 0.58	73 · 117 · 10
of 187 poisoned rounds restored, conflict vs random	poisoned labels found per four audits	seeds won, tied and lost by conflict auditing



Correlated decks, directed attack, four audits: once any labels are flipped, conflict auditing keeps the highest accuracy.



With 4 flips and 4 audits, the mean difference is never negative; the edge is largest where the attack bites hardest.

6 Where it fails

Conflict auditing needs a visible disagreement. The 187 poisoned rounds, split (exploratory):

After the attack	Rounds	Conflict	Random
Look-alikes disagree	78	87%	23%
No disagreement left	109	21%	9%

Flipped by the attacker (true rule: round)

C23 STORE · C19 STORE · C12 STORE · C15 STORE

Conflict audit: learned red (wrong)

1. C11 was right · 2. C13 was right · 3. C02 was right · 4. C05 was right

Random audit: learned round (restored)

1. C02 was right · 2. C21 was right · 3. C23 found flip · 4. C12 found flip

Seed 1040: every flipped tag sits in a look-alike group with no dissent, so conflict auditing had no signal and checked four correct cards.

7 What it means

Disagreement is a free signal worth auditing first, not proof of clean data: when poisoned look-alikes all agree, both policies mostly fail.

- Three rules, 24 cards, a perfect oracle at uniform cost.
- A heuristic attacker that ignores the defense; flipping whole look-alike groups could erase much of the edge.
- Simulated policies, not measured human play.

Play it, rerun it

github.com/haydn-s/poisoned-dataset-game

Rules, print files, raw results and the code behind every figure.

Online: `streamlit run streamlit_app.py`

Terminal: `python3 -m poisoned_dataset play`

Rerun: `python3 -m poisoned_dataset experiment`

Vassilev et al., *Adversarial Machine Learning: A Taxonomy and Terminology*, NIST AI 100-2e2025. Jha, Hayase and Oh, *Label Poisoning is All You Need*, NeurIPS 2023.